

Course Code	Course Name
2114112	Database Management System

Syllabus for Internal Assessment -I (Jan – Apr 26)

Subject: DBMS SE SEM IV

Branch: Computer

Div: - C1, C2, C3

1 Introduction to Database and Data Modeling

Introduction: Definitions and application, Characteristics of databases, DBMS architecture,

The Entity-Relationship (ER) Model: Entity types: Weak and strong entity sets, Entity sets, Types of Attributes, Keys, Relationship constraints: Cardinality and Participation, Extended Entity-Relationship (EER) Model: Generalization, Specialization and Aggregation

2. Relational Model and Relational Algebra

Introduction to the Relational Model, relational schema and concept of keys. Mapping the ER and EER Model to the Relational Model,

Relational Algebra-operators (Selection(σ), Projection(π), Union($\cup$), Difference($-$), Cartesian Product($\times$), intersection ($\cap$), Rename (ρ)), Relational Algebra Queries

3. Structured Query Language (SQL

Overview of SQL, Data Definition Commands, Integrity constraints: key constraints, Domain Constraints, Referential integrity, check constraints

Question for Reference for Internal Assessment -I

Q 1. Explain Characteristics of databases

Q 2. Explain Following with example

1. Entity types: Weak and strong entity sets, Entity sets
2. Types of Attributes
3. Keys, Relationship constraints
4. Cardinality and Participation
5. Extended Entity-Relationship (EER) Model: Generalization, Specialization and Aggregation
6. Total and Partial Participation
7. Ternary Relationship

Q 3. Construct ER Diagram on specified case study and convert to relational model.

Q 4. Explain Rules to Map the ER / EER Model to the Relational Model with example .

Q 5. Map given ER/EER model to relational model and also explain rule which is used mapping

Q 6. Explain following relational algebra operator with example

- a) Select
- b) Project
- c) Cartesian Product
- d) Union
- e) Difference ($-$)
- f) intersection ($\cap$)
- g) Rename

Q 7. Queries using Relational Algebra. Based on following operators

- b) Select
- b) Project
- c) Cartesian Product
- d) Union
- e) Difference($-$)
- f) intersection ($\cap$)
- g) Rename

Link for reference

1. <https://www.cs.emory.edu/~cheung/Courses/377/Syllabus/4-RelAlg/intro8.html>
2. <https://www.geeksforgeeks.org/dbms/introduction-of-relational-algebra-in-dbms/>
3. <https://www.cs.umb.edu/cs630/Lecture03.pdf>

Q 8 Explain **SQL** Data Definition Commands with example

Q 9. Explain SQL following Integrity constraints with example.

1. key constraints,
2. Domain Constraints

3. Referential integrity
4. check constraints

Link for reference

1. https://www.w3schools.com/sql/sql_check.asp

2. <https://www.geeksforgeeks.org/dbms/dbms-integrity-constraints/>

Subject Incharge

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